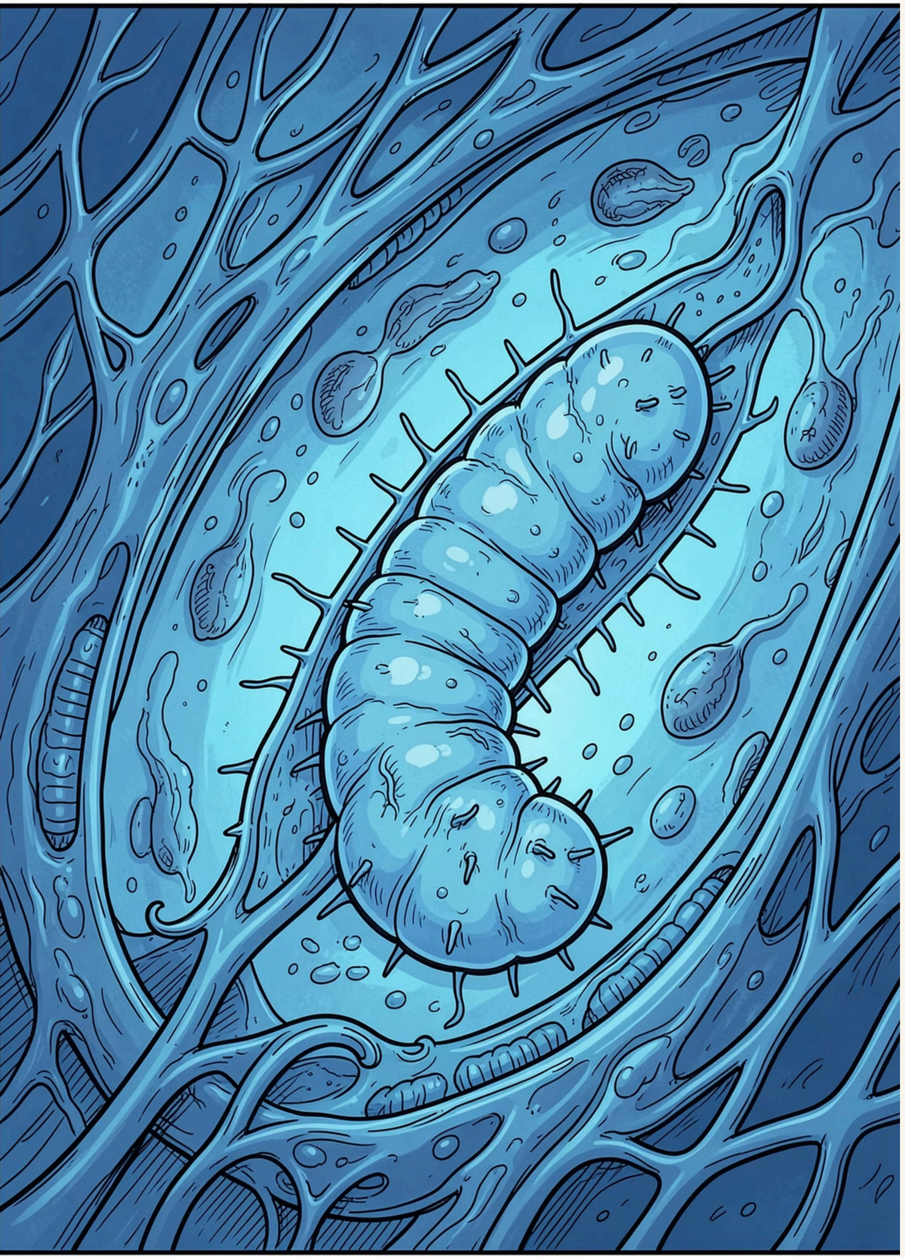
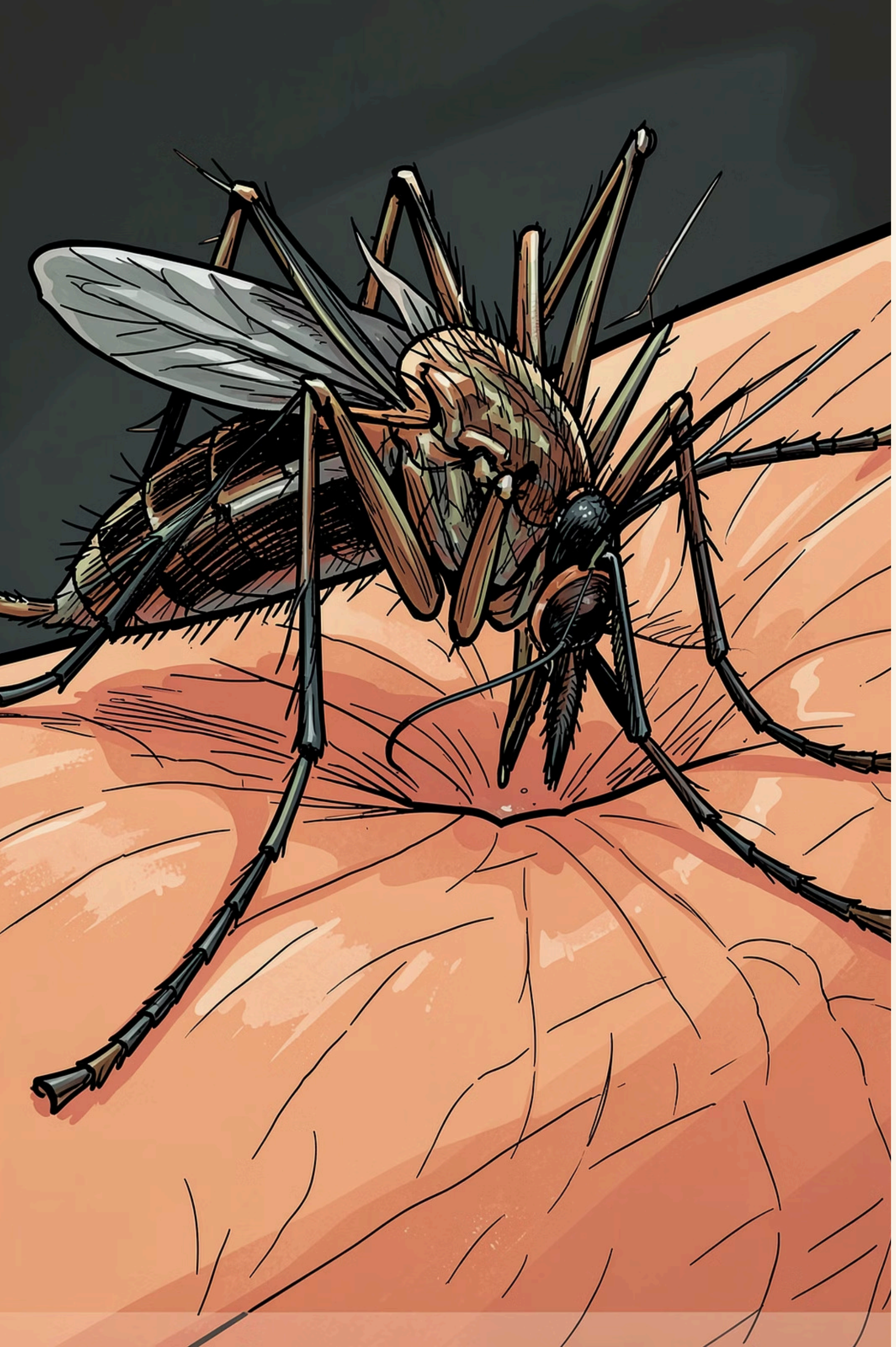


Wuchereria bancrofti

The parasitic nematode responsible for **Lymphatic Filariasis** — one of the world's most neglected tropical diseases, affecting millions across the tropics.

PARASITOLOGY · ZOOLOGY

NEMATODA · FILARIOIDEA



The Biological Cycle: A Two-Host Journey

Definitive Host — Humans

Adult worms inhabit lymphatic vessels and lymph nodes. Sexual reproduction occurs here, producing millions of **microfilariae** that circulate in blood.

Intermediate Host — Mosquitoes

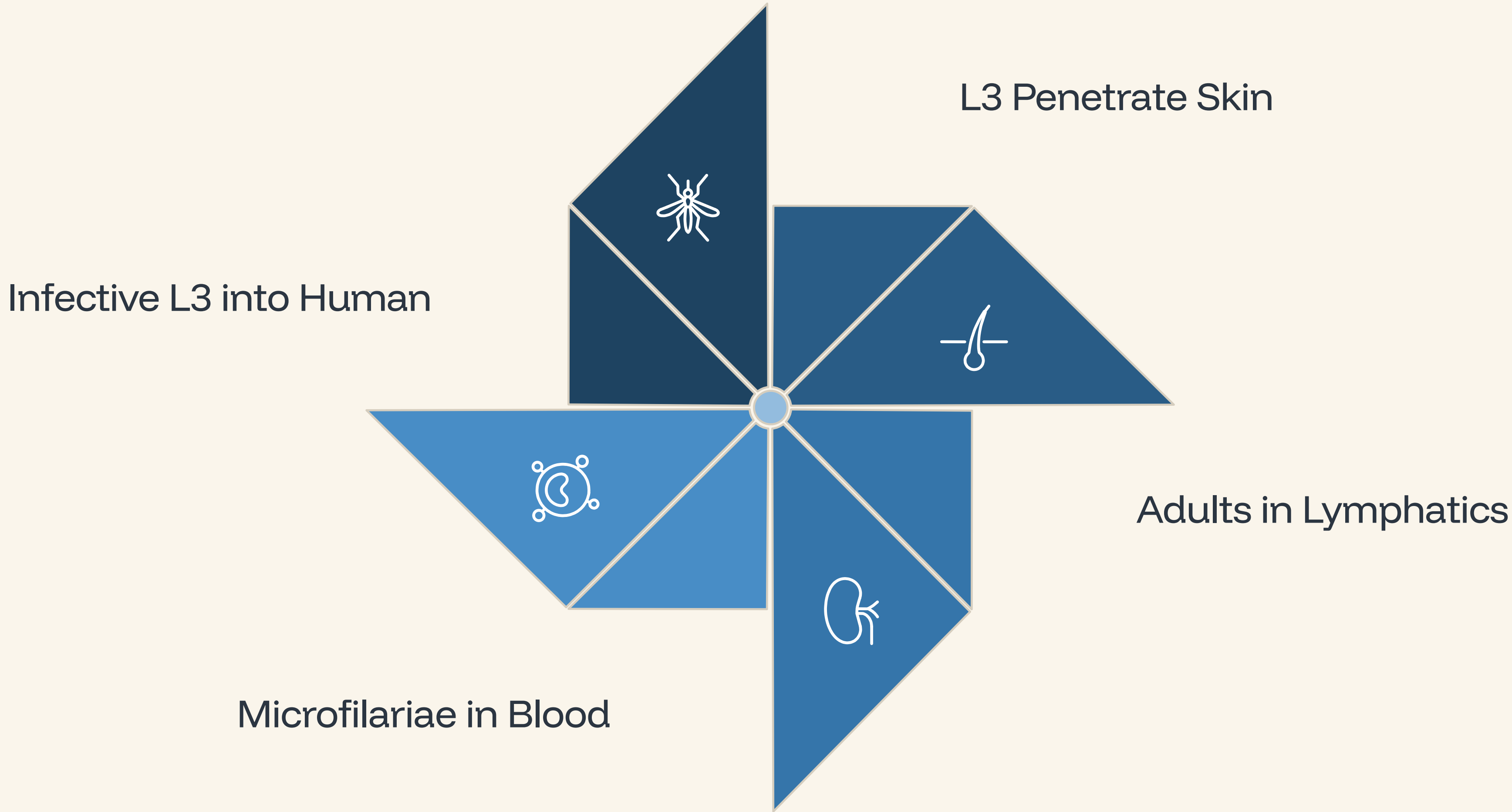
Genera **Anopheles**, **Culex**, and **Aedes** ingest microfilariae during blood meals. Larvae develop through L1–L3 stages in thoracic muscles.

Transmission & Maturation

Infective **L3 larvae** enter human skin via the mosquito bite wound, migrating to lymphatics and maturing into adults over **3 months to 1 year**.

The Complete Lifecycle of *W. bancrofti*

The parasite alternates between human and mosquito hosts in a tightly synchronized cycle driven by **nocturnal periodicity** — microfilariae peak in peripheral blood at night, coinciding with mosquito feeding activity.



Each stage is precisely timed — from larval penetration to adult reproduction — making the parasite highly adapted to its dual-host environment.

Pathogenesis & Clinical Manifestations

Immune-Mediated Damage

Pathology stems from the host's immune response to adult worms, triggering chronic lymphangitis and progressive fibrosis of lymphatic vessels.

Lymphatic Obstruction

Repeated inflammation causes calcified, non-functional vessels — blocking lymph drainage and causing fluid accumulation in tissues.

Visible Manifestations

Progresses from **lymphedema** → **hydrocele** (scrotal swelling) → severe **elephantiasis** of limbs and genitals.

Occult Filariasis

Tropical Pulmonary Eosinophilia — a hypersensitivity reaction with cough and wheezing, but *no detectable microfilariae* in blood.

Diagnosis, Treatment & Global Impact

120M+

Infected Individuals

Globally affected by lymphatic filariasis

1B+

People at Risk

Across tropical and subtropical regions

73

Endemic Countries

Requiring mass drug administration programs

Diagnosis

- Microscopic identification of **microfilariae** in night blood smears
- Circulating Filarial Antigen (CFA) tests — rapid and sensitive
- Ultrasound for detecting adult worms (*"filarial dance sign"*)

Treatment & Prevention

- **Diethylcarbamazine (DEC)** — primary antifilarial drug
- **Ivermectin + Albendazole** — combination therapy
- Mass Drug Administration (MDA) to break transmission
- Vector control and protective measures against mosquitoes